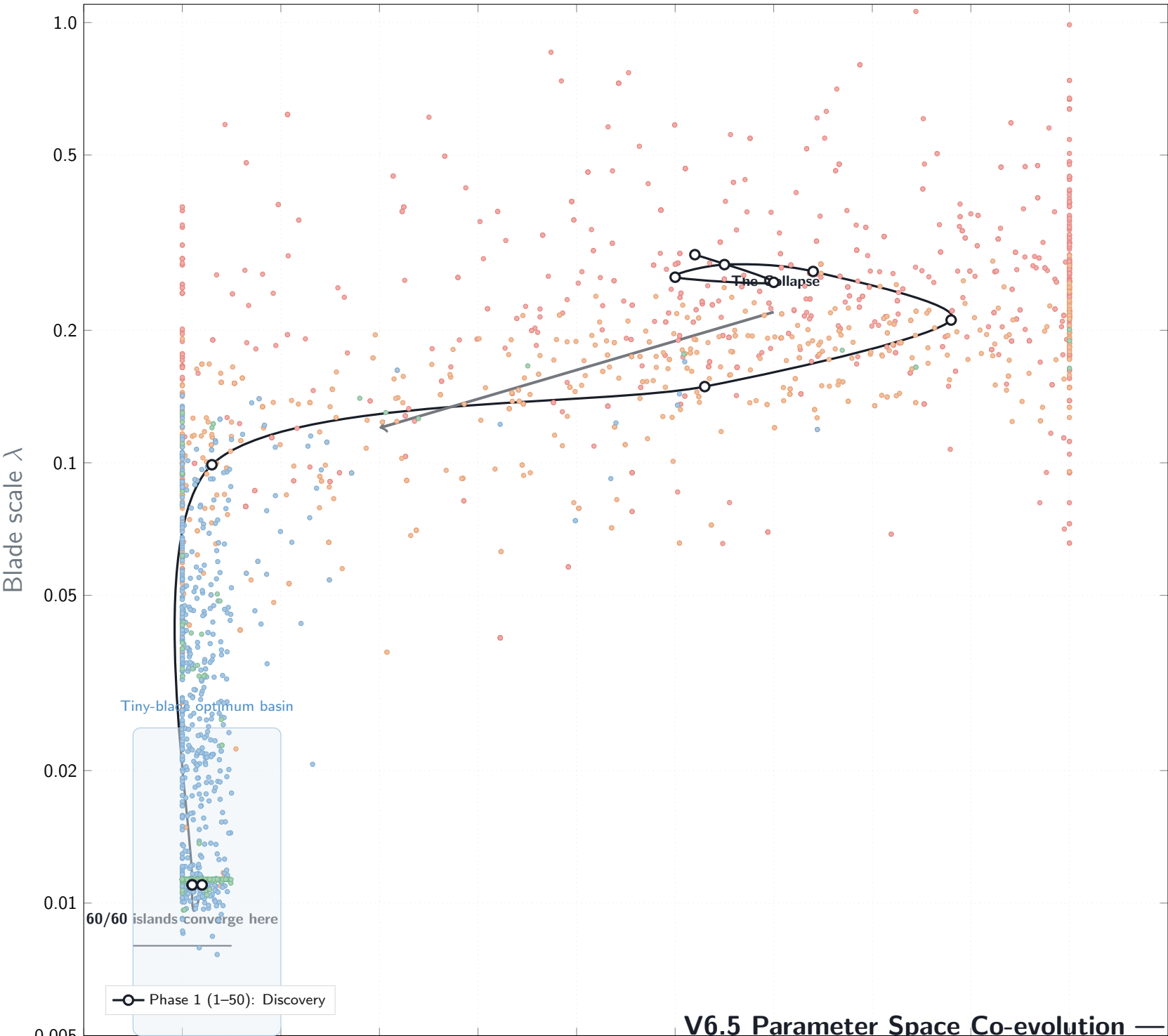


Co-evolution: Blade Scale λ vs Polygon n_{lines}

Each point = one island at one iteration. Color = convergence phase. Arrow = ensemble median trajectory through parameter space.



How to read this diagram:

Each colored dot shows where one island was at one iteration in (n_{lines}, λ) space.

Phase 1 (red): Wide exploration — all polygon sizes, $\lambda=0.1-0.8$. Scattered.

Phase 2 (orange): Population starts clustering around $n=8-11$ with moderate λ .

Phase 3 (blue): Abrupt collapse — n drops to 3, λ plummets to 0.01.

Phase 4 (green): Dense cluster at $(n=3, \lambda=0.011)$ — 60 islands unanimous.

The black path = ensemble median through parameter space across iterations.

Key insight: The optimizer spends Phase 1–2 exploring moderate polygons ($n=6-12$), then suddenly discovers that triangles + tiny blades ($\lambda=0.01$) unlock $10\times$ lower mass in Phase 3.